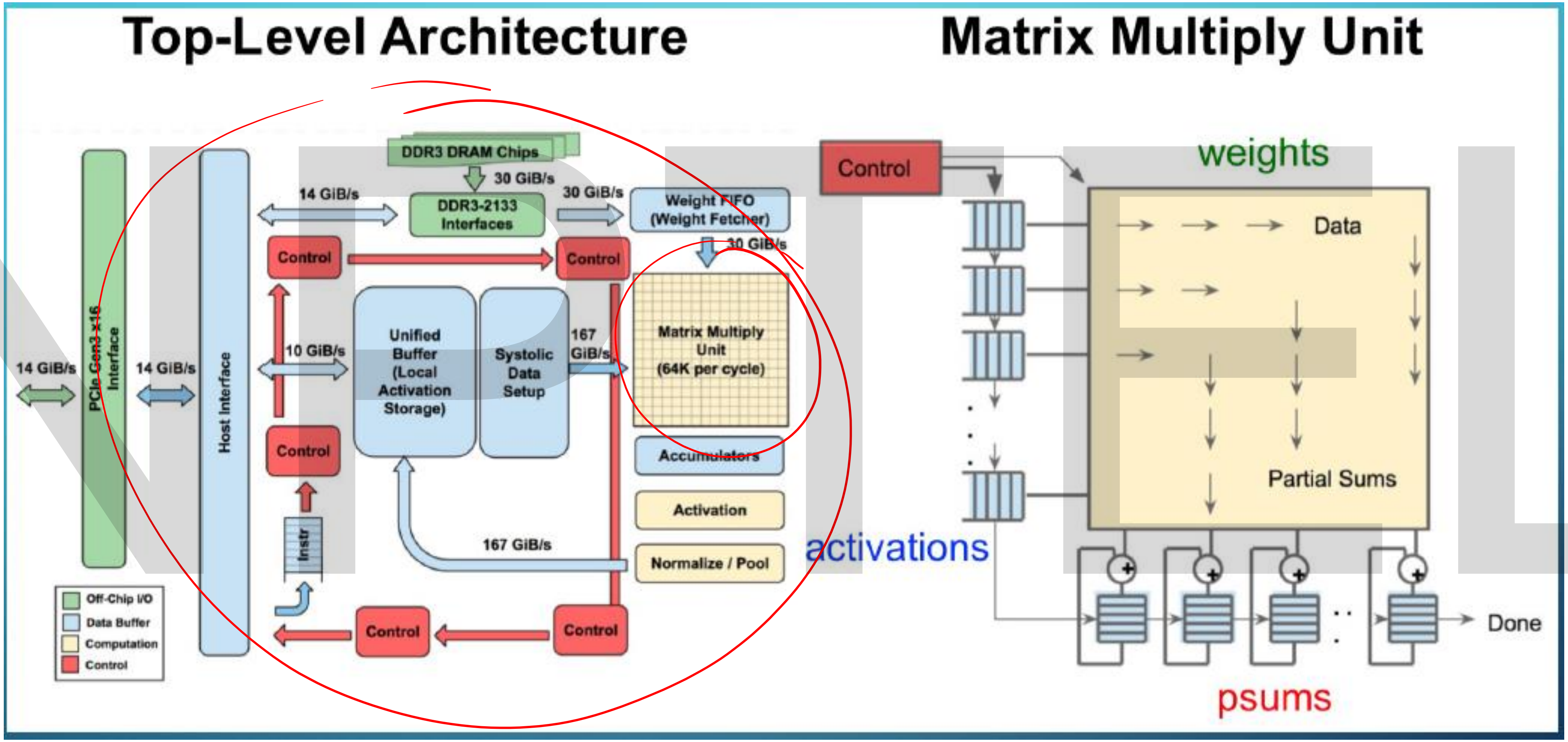
NPTEL Course on
"Computer System Optimization for ML"

Lecture #8

Agenda

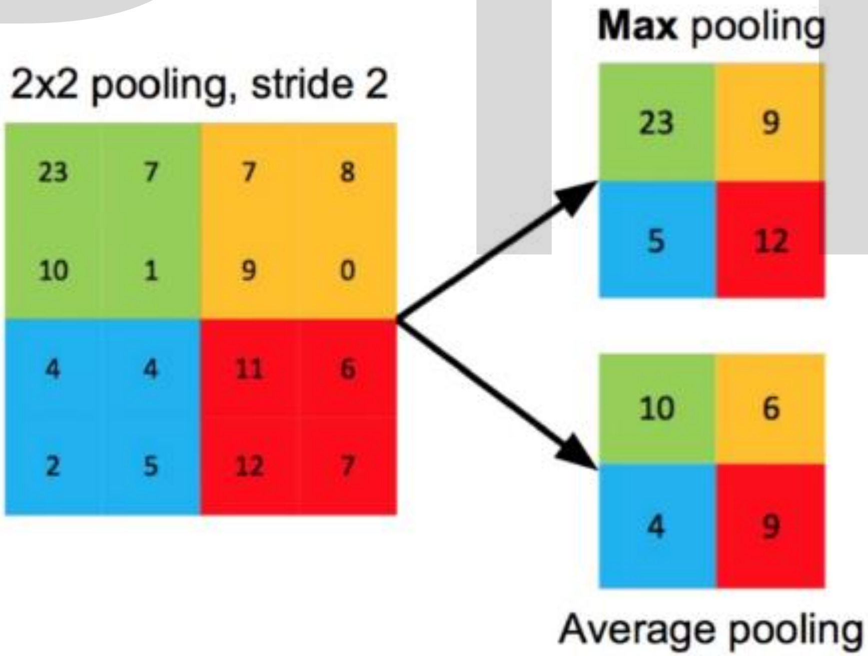
- Recap
- More Non-Linear Operations
- Popular CNNs

Tensor Processing Unit



Pooling (Pool) Layer

- **Reduce resolution of each channel independently**
- **Overlapping or non-overlapping**
  – Depends on stride
- **Increases translational-invariance and noise-resillience**



Agenda

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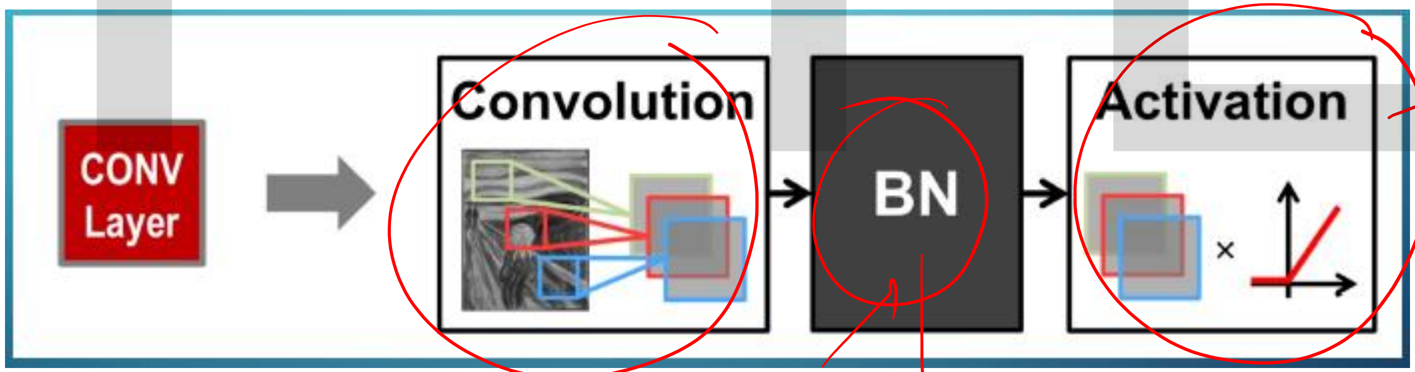
Normalization Layer

- **Batch Normalization**
  - Normalization activations towards mean=0 and std dev=1 based on the statistics of the training data set
  - Put between conv/FC and activation function
- **Believed to be key to getting high accuracy and faster training for DNNs**



ReLU, sigmoid.

Normalization Layer

- **The normalized values are further scaled and shifted**
  - The parameters are learnt through training

data mean

learned scale factor

mean

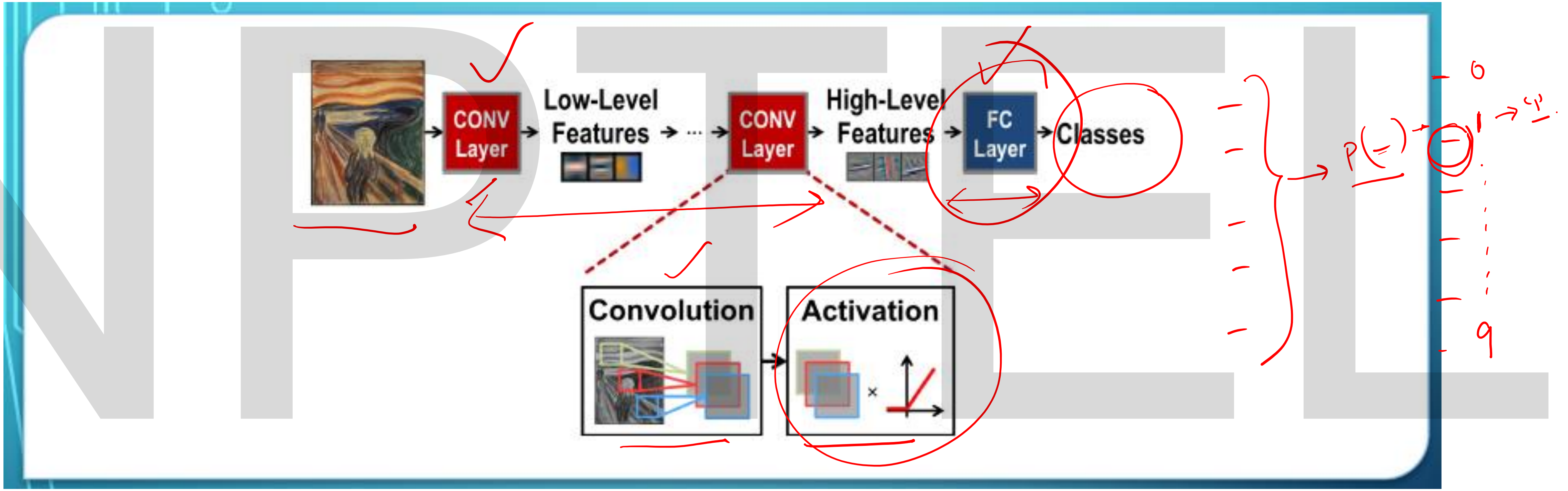
$$y = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}}\gamma + \beta$$

learned shift factor

data std. dev.

small const. to avoid numerical problems

Deep CNNs (stitching everything together)



Agenda

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- **Popular CNNs**

Commonly used Dataset

- **MNIST**
  - Digit Classification
  - 28×28 pixels (B&W)
  - 10 Classes
  - 60,000 training
  - 10,000 testing

28

28

0 . . . 9

28

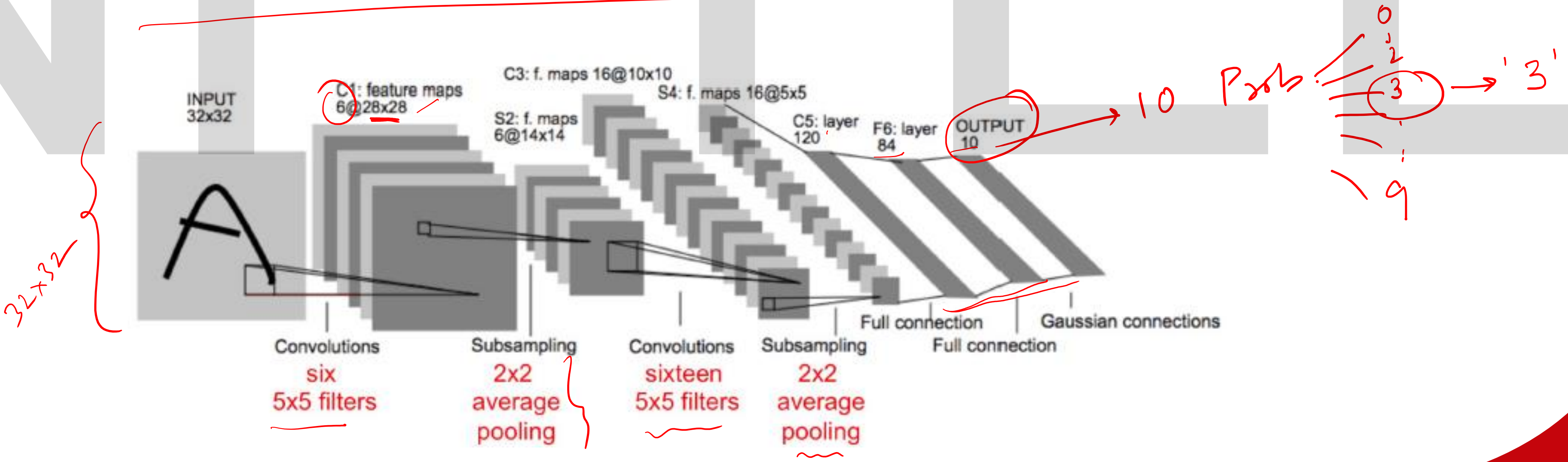
28

7

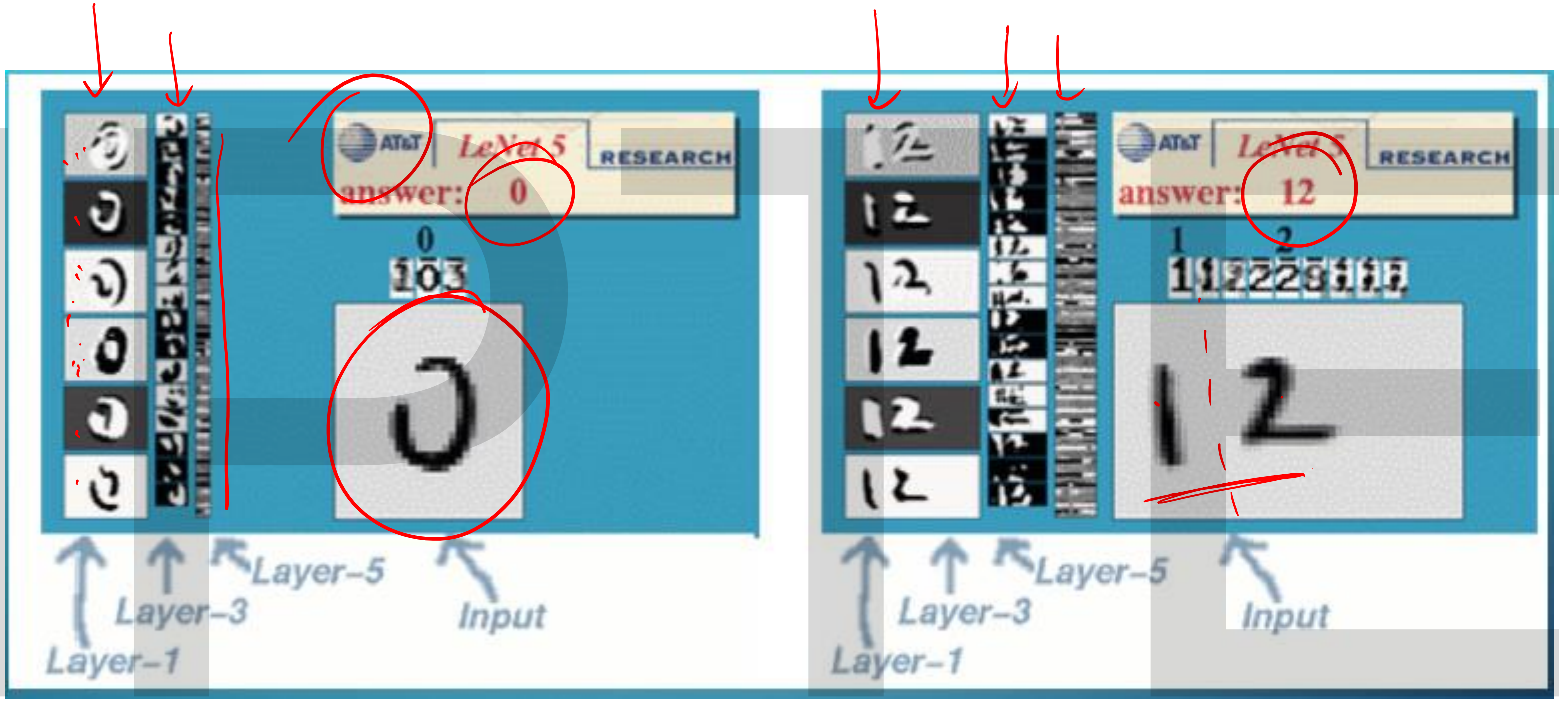
3

LeNet-5

- Conv layers: 2
- Fully connected layers: 2
- Weights: 60k
- MACs: 341k
- Sigmoid used for non-linearity



LeNet-5



AT&T
LeNet 5
RESEARCH
answer: 0
Layer-1
Layer-3
Layer-5
Input
AT&T
LeNet 5
RESEARCH
answer: 12
Layer-1
Layer-3
Layer-5
Input

ImageNet

- **Image classification**
  - 256×256
  - Colour images
  - 1000 classes
  - 1.3 M training
  - 100, 000 testing
- **For ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**
  - Accuracy of classification task reported based on top-1/top-5 error

1000 pr(-)

[-]

ImageNet

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  - Colour images
  - 1000 classes
  - 1.3 M training
  - 100, 000 testing

- **For ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**
  - Accuracy of classification task reported based on top-1/top-5 error
  - What is top-K error?

AlexNet (Krizhevsky et al., NeurIPS 2012)

- **ILSCVR12 Winner**
- **Uses local response normalization (LRN)**
- **Structure**
  - 5 conv layers
  - 3 fully connected layers
  - Weights: 61M
  - MACs 724M
  - ReLU used for non-linearity

AlexNet (Krizhevsky et al., NeurIPS 2012)

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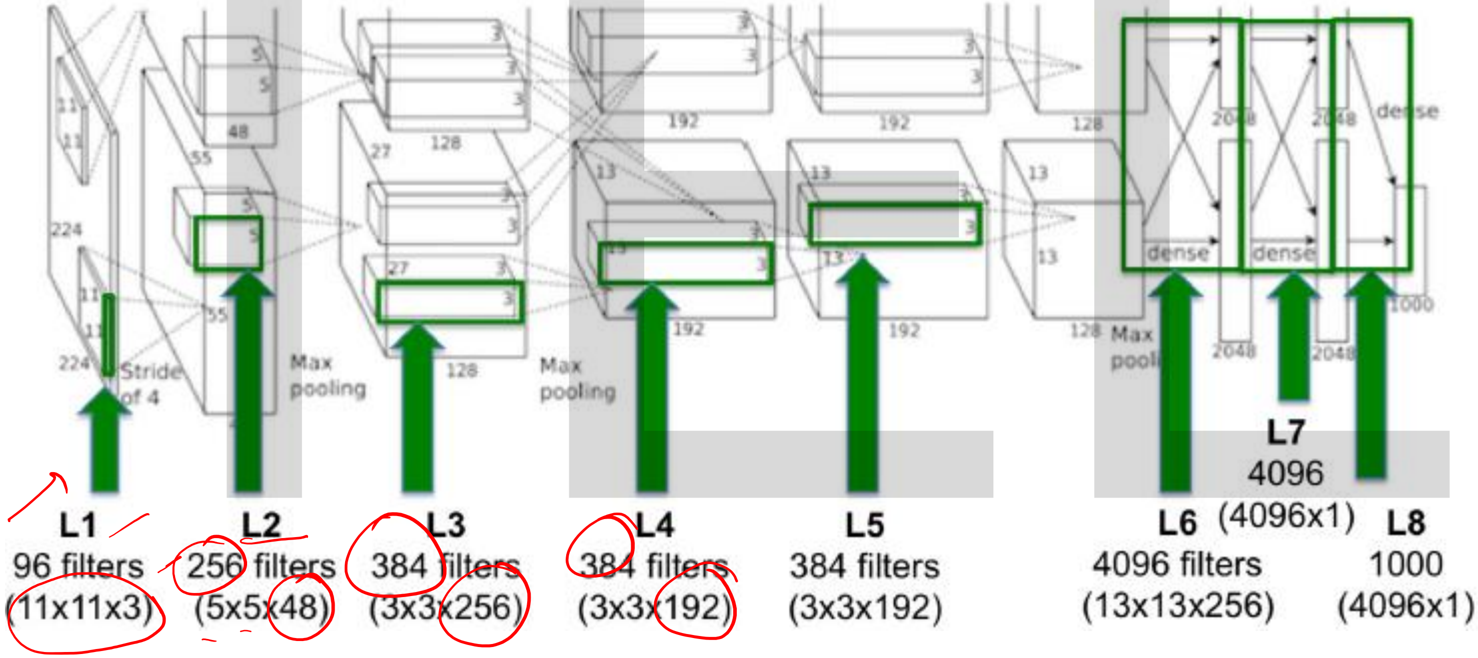
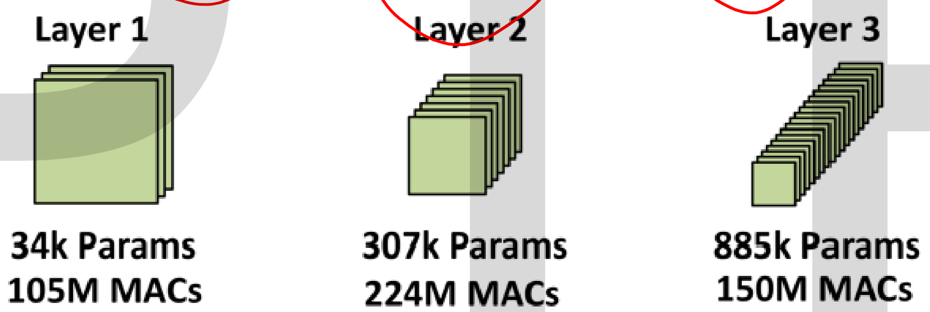
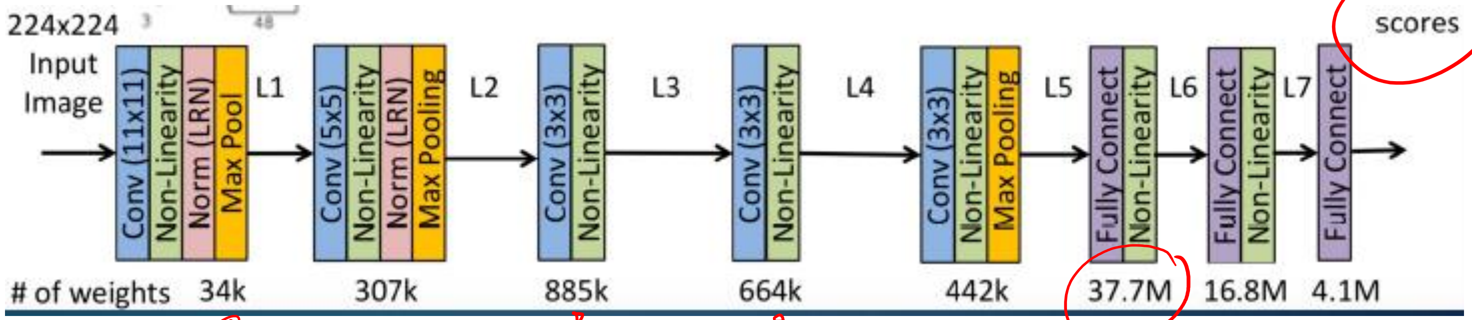


AlexNet: Large Sizes with Varying Shapes

AlexNet Convolutional Layer Configurations

Layer	Filter Size (RxS)	# Filters (M)	# Channels (C)	Stride
1	11x11	96	3	4
2	5x5	256	48	1
3	3x3	384	256	1
4	3x3	384	192	1
5	3x3	256	192	1





VGG-16 (Simonyan et al., ICLR 2015)

- Conv layers: 13
- FC layers: 3
- Weights: 138M
- MACs: 15.5G
- There is a 19-layer version too (VGG-19)

→ 16

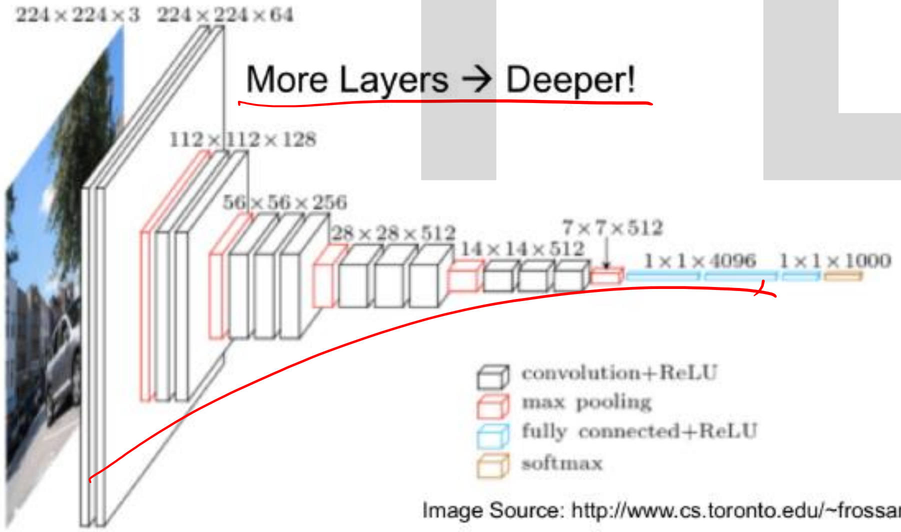
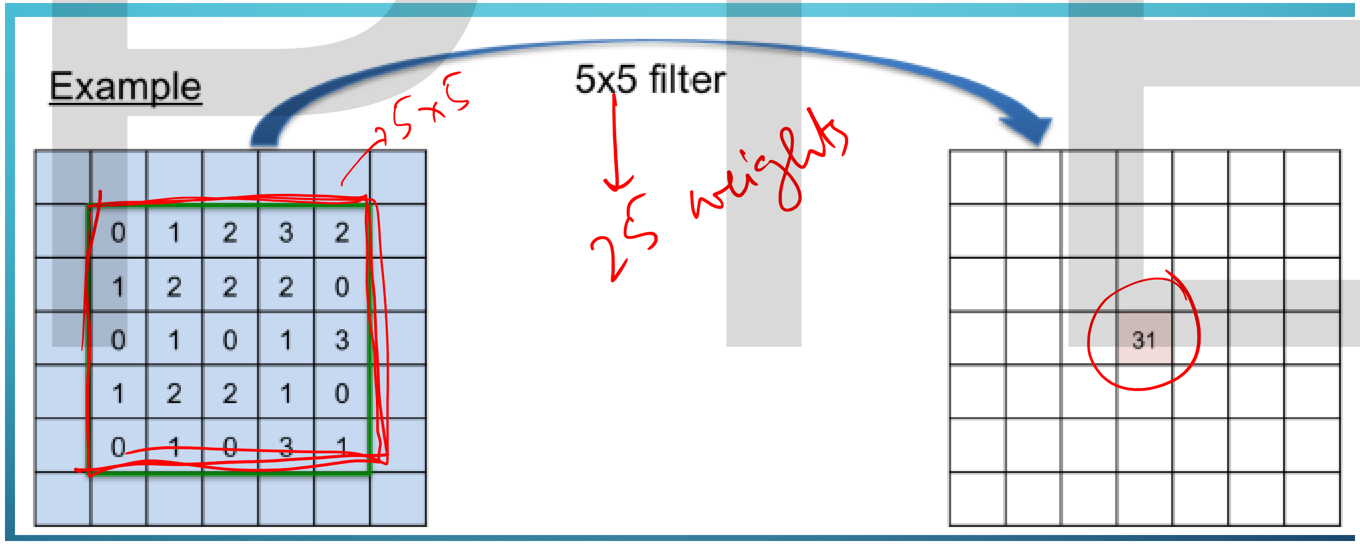


Image Source: http://www.cs.toronto.edu/~frossard/post/vgg16/

Stacked Filters

- Deeper networks means more weights
- Use stack of smaller filters (3×3) to cover the same receptive field with fewer filter weights



Stacked Filters

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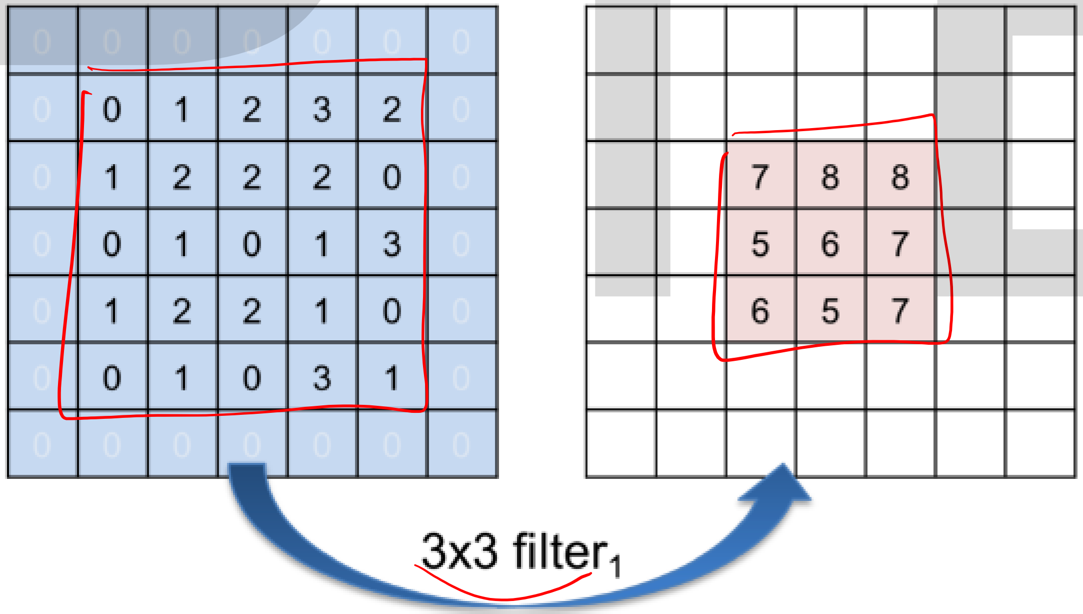
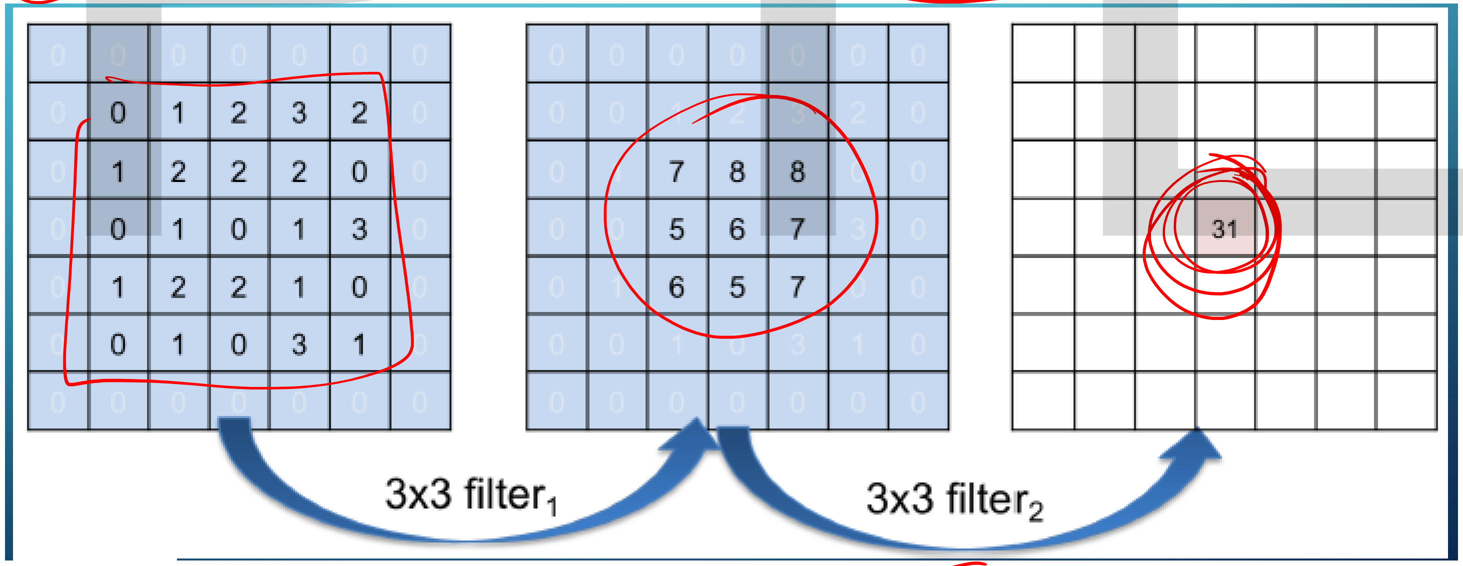
Example



Stacked Filters

- Deeper networks means more weights
- Use stack of smaller filters (3×3) to cover the same receptive field with fewer filter weights
- Non-linear activations inserted between each filter
  - 5×5 filter (25 weights) → two 3×3 filters (18 weights)



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)